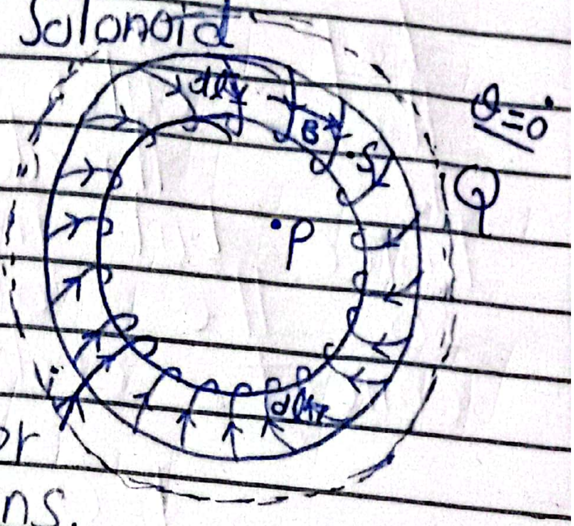
Magnetic field due to Toroid

Toroid is a Circular Solonoid

OR

Toroid is a coil of insulated copper wound on a circular cylinder with closed turns.



Using Ampere's Circuital law

$$\oint B.dl = \mu_0 \times I$$

Magnetic field at Point 'P' (Inside)

$$\oint B.dl = \mu_0 I_{inside}$$

$$I_{inside} = 0$$

$$\oint B.dl = 0$$

$$\boxed{B = 0}$$

Magnetic field at 'Q' (outside)

Loop on outside the core on the outer side of the toroid is threated by each turn twice but in opposite direction so the algebric sum of the current enclosed is zero

$$I = 0$$

So magnetic field is again zero

$$\boxed{B = 0}$$

Magnetic field at Point 'S'

$$\oint B.dl = \mu_0 I$$

$$\oint B dl \cos\theta = \mu_0 I$$

$$\oint B dl \cos 0^\circ = \mu_0 I$$

$$B \oint dl = \mu_0 I$$

$$B 2\pi r = \mu_0 I$$

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Total Number of turn in toroid is N, So on each turn the current is I and for N turn will be NI

$$\boxed{B = \frac{\mu_0 N I}{2\pi r}}$$

For an ideal Toroid

$d <<< R$

Let 'n' be number of turn per unit length

$$n = \frac{N}{2\pi r}$$

$$B = \frac{\mu_0 n \times \cancel{2\pi r} \times I}{\cancel{2\pi r}}$$

$$\boxed{B = \mu_0 n I}$$